ie UNIVERSITY | SCHOOL OF SCIENCE & TECHNOLOGY

BACHELOR IN DATA AND BUSINESS ANALYTICS

TIME SERIES ANALYSIS

Session 2 · Time Series Graphics

Professor: Angel Mora Segura

03 September 2026 · 18:00–19:30 · Live in-person



COURSE SCHEDULE · 20 SESSIONS · 80 MINUTES EACH

Session by Session

1	02 Sep	Introduction · R setup · forecasting as a business problem ✓
2	03 Sep	Time plots, seasonal plots, EDA **TODAY**
3	23 Sep	Lags, ACF, PACF
4	24 Sep	Decomposition, STL, Box-Cox
5	30 Sep	Benchmarks: Naive & Seasonal Naive
6	01 Oct	Simple exponential smoothing + Holt
7	07 Oct	Holt-Winters / ETS
8	08 Oct	ETS model selection + validation
9	15 Oct	**Intermediate Test — 20% · Checkpoint 1**
10	19 Oct	AR / MA
11	22 Oct	ARMA + stationarity
12	28 Oct	ARMA + integration + diagnostics
13	29 Oct	ARIMA
14	12 Nov	Seasonal ARIMA (SARIMA)
15	18 Nov	Automatic ARIMA + model selection
16	19 Nov	Complex seasonality / Fourier · Checkpoint 2
17	25 Nov	Integrated Forecasting Workshop
18	26 Nov	Business Case review & feedback
19	03 Dec	**Group presentations — 30%**
20	11 Dec	**Final exam in R — 40%**

WHERE WE LEFT OFF

What Session 1 Established

Forecasting is a business problem first

You wrote a one-sentence brief: what variable, at what granularity, how often, how far ahead, for whom, to decide what. Everything today is judged against it.

Three patterns to name

Trend, seasonality and cycles — plus outliers and changing variance. Session 1 defined them. Today you learn to **see** them.

The five-step workflow

Problem → data → **exploratory analysis** → model → evaluate. Today is step 3, and it is where model selection really begins.

Your case is running

Each team owns one of Happy Cow, Balearic Airlines, Clean Air or Wayfair for the whole term, and logs every decision as it is taken.

If your data is not yet in a tsibble, that is today's first lab task — you cannot plot what you cannot index.

TODAY'S SESSION

Seeing the Data

Time plots, seasonal plots, subseries plots and scatterplots — the four graphics that decide which models you are allowed to use.

FPP3 · Chapter 2

TYPICAL CLASS STRUCTURE · 80 MINUTES

The Classroom Is the Laboratory

The same three blocks every session. The work on your case happens here, with me in the room.

~30 MIN

BLOCK 1

Concepts & R application

The methodology, its intuition and its assumptions — aimed at **what problem this technique solves** — then straight into its practical implementation in R.

You leave with code that runs.

~30 MIN

BLOCK 2

Business Case Lab

Teams apply or evaluate the methodology in the context of their own case. Finished code is not the requirement.

Output can be a plot, a methodological decision, a hypothesis, a data transformation, a fitted model, or a justified argument for why this technique does not apply.

~20 MIN

BLOCK 3

Business Case Clinic

Selected teams share their approach, findings or difficulties in about 5 minutes each.

The class discusses alternatives and gives feedback; I close the loop.

Block durations vary with the topic: technically demanding sessions take more time for explanation and R; later sessions shift progressively towards project development and feedback.

BY THE END OF TODAY YOU SHOULD BE ABLE TO

Session Objectives

1. Build a **time plot** that is actually readable: right frequency, right aspect ratio, grid resolution that lets you read values off it.

2. Use **seasonal plots** and **seasonal subseries plots** to detect a seasonal pattern, its shape, and whether it is changing over time.

3. Use **scatterplots** and **lag plots** to examine relationships between series, and recognise when correlation is the wrong summary.

4. Produce and interpret this set of plots **for your own case**, and write down what each one rules in or out.

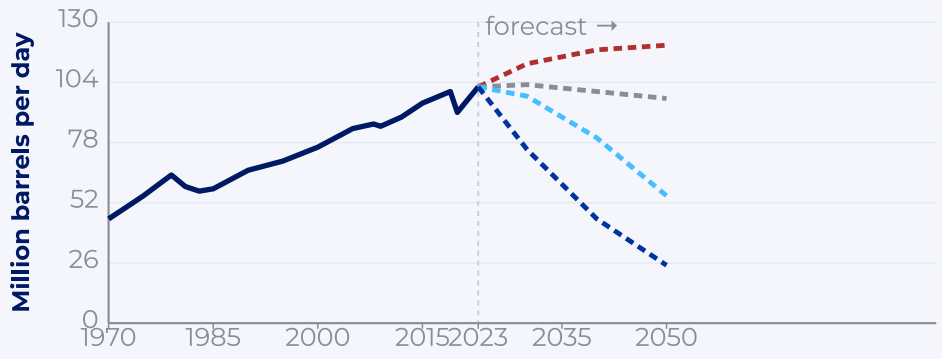
BLOCK 1 · 30 MINUTES

Concept + R Application

Four graphics, in the order you should always produce them.

CONTEXT THE ECONOMIST · SPECIAL REPORT · “THE END OF OIL, THEN AND NOW” · MARCH 2024

The Most Consequential Forecast in the World, and Nobody Agrees on It



Schematic reconstruction for teaching purposes, based on the scenarios discussed in the article. Not the published figures.

One history. Four futures. All produced by serious institutions with access to the same data.

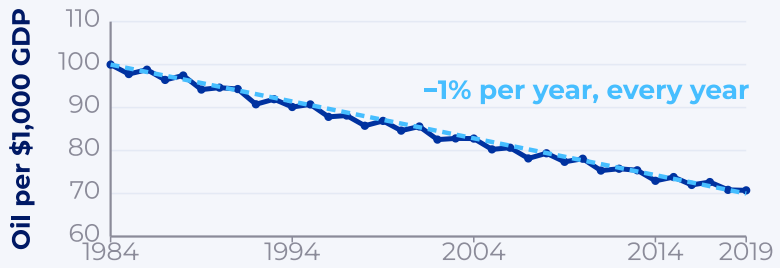
- In 1977 Jimmy Carter told Americans that falling US production was a sign of reserves running out everywhere. The premise was wrong: Alaskan oil and, later, hydraulic fracturing changed the supply picture entirely.
- The policy response worked anyway. US oil consumption fell 17% between 1977 and 1985 while GDP rose 27% — and the resulting glut nearly broke OPEC.
- Today the IEA sees demand falling by a quarter by 2030 under aggressive policy; OPEC and ExxonMobil see it growing into the 2030s. Same series, opposite conclusions.

“We are quite good planning the supply but always wrong on demand.”

Patrick Pouyanné, CEO of TotalEnergies

WHY THIS BELONGS IN A SESSION ABOUT GRAPHICS

Four Lessons for Reading Any Time Series



The oil needed per $1,000 of output fell by almost exactly 1% a year from 1984 to 2019 — a trend nobody legislated and nobody noticed year to year.

A trend is invisible up close. One per cent a year is inside the noise of any single year, and transforms the system over thirty-five. Only the plot makes it visible — and only if the axis is scaled to show it.

Falling intensity is not falling demand. Efficiency improved while consumption grew. Two series that both matter, pointing in opposite directions — a single chart of either one, alone, misleads.

The forecast fan *is* the finding. When credible scenarios span 25 to 120 mb/d, the honest deliverable is the spread and its assumptions, not a single line down the middle.

Structural breaks beat models. Fracking, the 1985 price collapse, COVID. Each is a break you can see instantly on a time plot and that no extrapolation of the prior data would have produced.

At an IEA meeting in 1983, expert forecasts of oil demand in 2000 did not merely disagree — in several cases the entire range missed. They had underestimated efficiency gains already under way, and it was visible in the data.

Why We Always Plot First

Graphs are the first thing to do with any time series, and often the fastest route to the answer. A summary statistic collapses a series into a number; a plot preserves the order, and order is where the information lives.

Plots reveal what tables hide

Patterns, unusual observations, changes over time and relationships between variables are visible in seconds and invisible in a column of numbers.

Every feature you see becomes a modelling requirement

Trend → the model needs a trend component. Seasonality → seasonal terms and a seasonal benchmark. Variance that grows with level → a transformation.

The client reads plots, not coefficients

In a consulting engagement the first deliverable that earns trust is a clear picture of their own business. You will show these plots in your final presentation.

The type of data you have determines which graph to use — and how the data is indexed determines what R will even let you draw.

EXAMPLE HAPPY COW · MONTHLY UNITS, THREE YEARS

The Time Plot



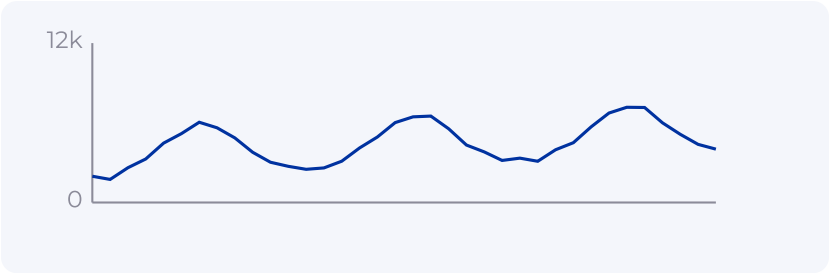
Observations plotted against time, joined by lines.

- The line matters: it asserts that consecutive points are related. Never join points that are not consecutive in time.
- **Trend:** each summer peak is higher than the last — the level is drifting upward.
- **Seasonality:** a peak every July, a trough every January. Three repetitions of the same shape.
- **Amplitude:** the swing is roughly constant in absolute units, so an additive model is plausible. If it grew with the level, we would transform.
- Dashed lines mark year boundaries — free orientation for the reader, at no cost.

Make the Plot Readable: Resolution and Aspect Ratio

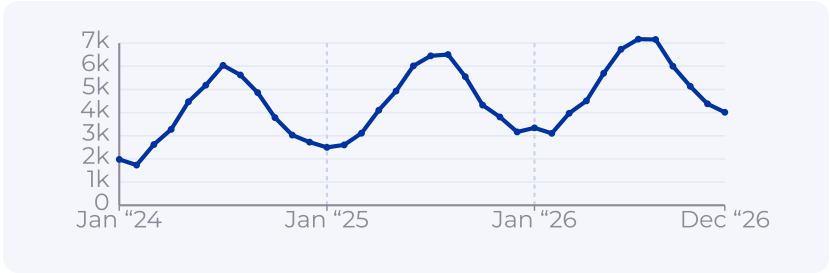
The same data can hide or reveal its pattern depending on how you draw it. Three settings do most of the work: the vertical range, the density of the gridlines, and the width-to-height ratio.

Hard to read



Axis stretched to 12k, no gridlines, squashed vertically. The seasonal swing looks like noise and no value can be read off the chart.

Readable



Range fitted to the data, gridlines every 1,000 units, year boundaries marked, points visible. Now the trend and the seasonal shape are both obvious.

Rule of thumb: choose gridlines so a reader can estimate any value to within one gridline, and set the aspect ratio so the average slope of the interesting segments is near 45°.

R APPLICATION

Drawing a Time Plot

```
library(fpp3)
# 1. the default time plot
sales |> autoplot(units)
# 2. controlled and labelled
sales |>
  autoplot(units) +
  labs(title = "Happy Cow: monthly units",
       y = "Units sold", x = NULL) +
  scale_y_continuous(
    breaks = seq(0, 7000, 1000)) +
  theme_minimal()
```

`autoplot()` reads the index of the tsibble, so the x axis is correct by construction — that is the payoff for building the tsibble properly.

Useful variations

- **Several series at once:** if the tsibble has a key, `autoplot()` draws one coloured line per key. Beyond about eight, facet instead of colour.
- **Zoom in:** `filter_index("2025 Jan" ~ "2025 Dec")` to inspect one stretch without rebuilding the data.
- **Change the frequency:** `index_by()` + `summarise()` to move from daily to weekly or monthly. Aggregation is a modelling decision — log it.
- **Check the gaps first:** `has_gaps()`, then `fill_gaps()`. A line drawn across a missing month is a lie.

What You Are Reading For

Go through this list every time, in this order, and write the answer down. Each one closes off part of the modelling space.

Level and trend

Where does the series sit, and is it drifting? A trend forces a trended method or differencing.

Repeating shape

Does the same pattern recur at a fixed period? If yes, seasonal naive becomes your benchmark.

Changing variance

Do the swings grow with the level? Then a log or Box-Cox transformation before anything else.

Outliers and breaks

Can you explain each one? A promotion, a strike, COVID. Unexplained spikes are data problems until proven otherwise.

Flat stretches and zeros

Stockouts, closures, or a sensor that stopped. Zero sales and no data are different things.

Enough history?

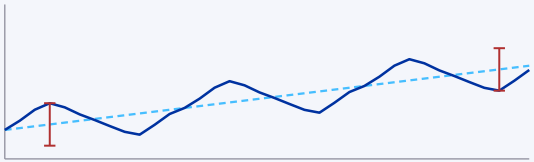
Count the complete seasonal cycles. Two is the bare minimum to estimate seasonality; fewer means you borrow it from elsewhere.

Each answer belongs in the decision log with one line of consequence: “variance grows with level → we will model log(units)”.

One More Thing to Read Off the Time Plot

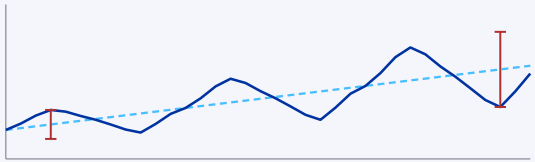
Compare the **width of the seasonal swing** with the **level of the trend**. That single comparison decides how the series should be modelled — and you make it by eye, before any code.

Additive



Same width at the start and at the end: seasonality is **independent of the level**. Model it as a quantity added to the trend — "+500 units every July".

Multiplicative



The swing grows with the trend: seasonality is **proportional to the level**. Model it as a factor — "July is always 40% above trend".

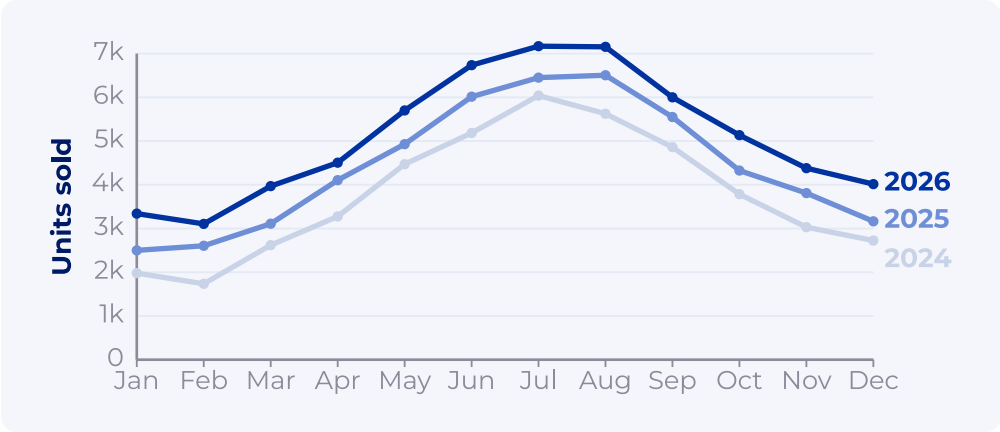
Why it matters now

- A multiplicative series modelled additively under-forecasts the peaks as the business grows — exactly when a stockout costs most.
- The standard fix is a **log or Box-Cox transformation**, which turns a multiplicative pattern into an additive one. That is Session 4.
- Growing businesses are usually multiplicative, mature ones additive. State which yours is, and why, in the log.

The classic sales-forecasting decision from the business literature — and it costs nothing, because you are already looking at the plot.

EXAMPLE SAME DATA, PLOTTED AGAINST THE SEASON

The Seasonal Plot



The same observations, but each year drawn as its own line against the season.

- **The shape is confirmed:** the three lines are near-parallel, so the seasonal pattern is stable — the same months are always high and low.
- **The peak is located exactly:** July, every year. A time plot suggests it; a seasonal plot pins it down.
- **The trend appears as separation:** the lines sit above one another. Vertical gaps are year-on-year growth.
- **Anomalies stand out:** a year that crosses the others is a year where the seasonal pattern broke — ask why.

R APPLICATION

gg_season()

```
sales |> gg_season(units)
# polar version: the year as a circle
sales |> gg_season(units,
                   polar = TRUE)
# pick the seasonal period explicitly
calls |> gg_season(volume,
                   period = "day")
calls |> gg_season(volume,
                   period = "week")
```

Colour encodes the year, so a legend that runs light-to-dark reads as time passing.

Series can have more than one season

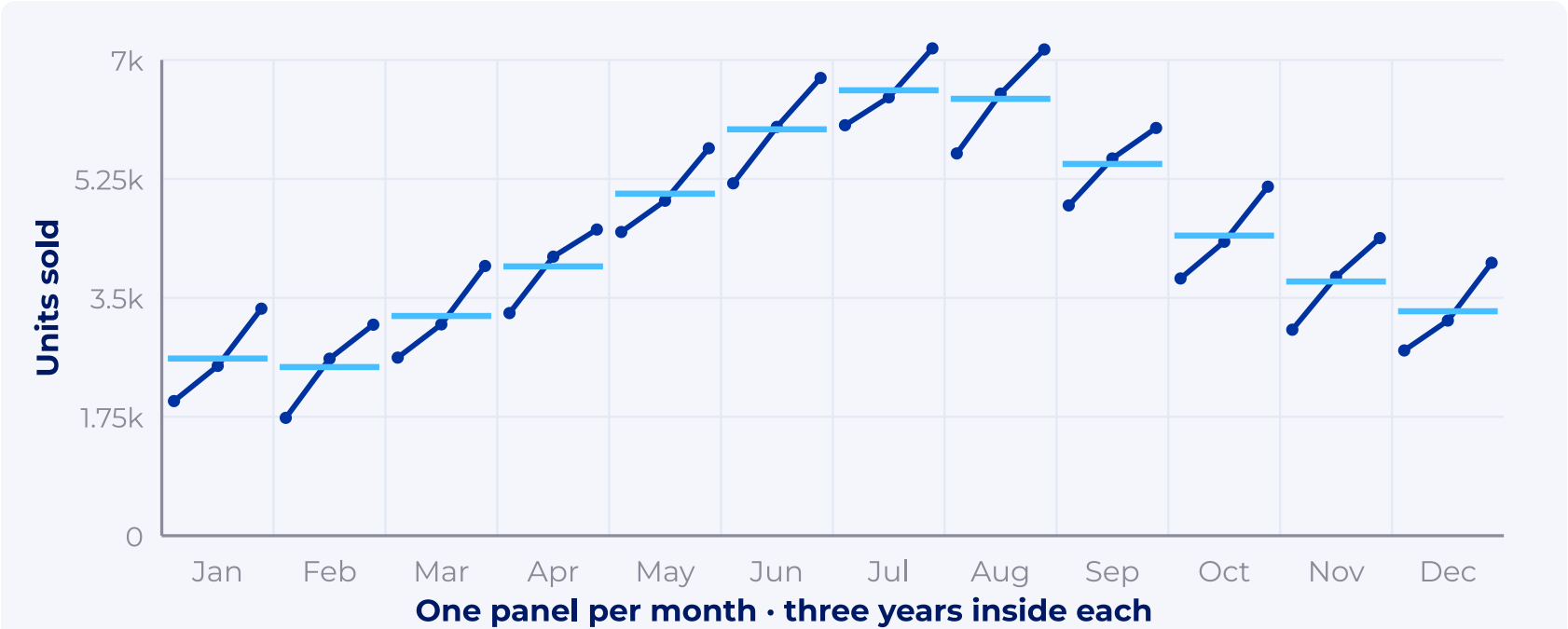
High-frequency data usually carries several seasonal periods at once, and each needs its own plot before you can claim to have seen the data.

- **Hourly call volumes:** a time-of-day pattern, a day-of-week pattern, and an annual pattern — three periods in one series.
- **Daily retail sales:** day-of-week and time-of-year, plus moving holidays that do not sit on a fixed date.
- **Monthly demand:** usually one annual period — simpler, but you need at least two full years to see it.

For your case: list every seasonal period you believe exists, then produce one `gg_season()` per period to confirm or kill it.

EXAMPLE THE SEASONAL SUBSERIES PLOT

One Panel per Season



How to read it: the data is cut into twelve mini time plots, one per month. Inside each panel time still runs left to right — 2024, 2025, 2026.

The blue horizontal line is that month's mean. Comparing those twelve lines gives you the seasonal shape, cleanly separated from the trend.

The slope inside a panel is the trend *for that season*. A rising July with a flat January means the summer peak is getting sharper — something no other plot shows.

R APPLICATION

gg_subseries() — and Choosing Between the Three

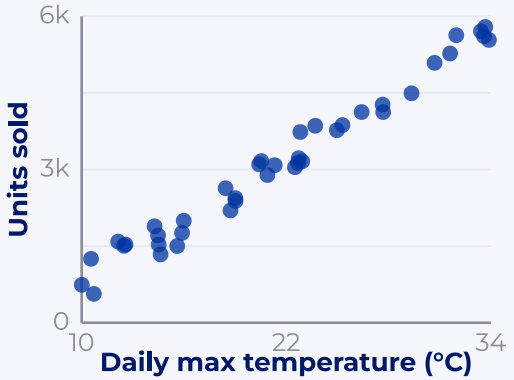
```
sales |> gg_subseries(units)
# with a key, one row of panels
# per series
sales |>
  gg_subseries(units) +
  labs(y = "Units sold")
```

Watch for: a panel whose blue mean line is far from its neighbours is a season that dominates the year — and usually the one the business cares about.

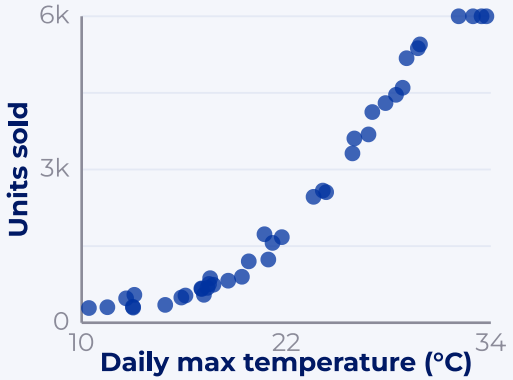
Time plot	Always first. Shows trend, breaks, outliers and whether the series is even usable.
Seasonal plot	When you suspect a seasonal period. Shows its shape, where the peak sits, and which years broke the pattern.
Subseries plot	When you need the trend *within* each season, or the seasonal means side by side. Best for “is our seasonality changing?”
All three	They answer different questions and take one line of code each. Produce all three for your case, every time.

Scatterplots: Relationships Between Series

Time plots show one series against time. Scatterplots show two series against **each other** — the first step towards a model that uses external information, and the way you decide whether a driver is worth collecting.



Strong, positive, roughly linear. Temperature is worth collecting — but only if you can obtain it for the forecast period.



Strong but non-linear. Almost nothing happens below 20°C, then demand accelerates. A straight line fitted here is wrong at both ends.

What to check

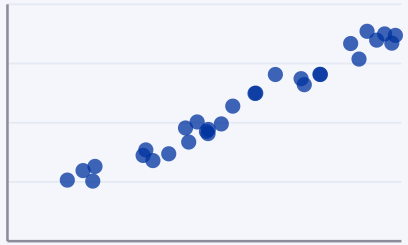
- **Direction and strength** — and whether it is strong enough to justify the extra data pipeline.
- **Shape** — linear, curved, or with a threshold. Shape dictates whether you need a transformation or a spline.
- **Outliers** — a single extreme point can create or destroy an apparent relationship.
- **Availability at forecast time** — a perfect predictor you will not know in advance is useless for forecasting.

Correlation between two time series is often driven by a trend both of them share. Two series can move together for years and have no causal link whatsoever.

Correlation Is Not the Relationship

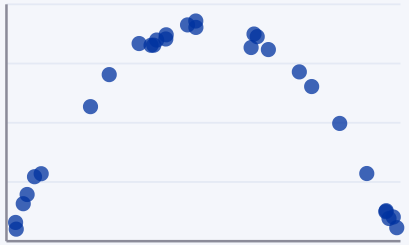
The correlation coefficient measures the strength of a **linear** relationship, and nothing else. These three datasets are why you look at the scatterplot before you quote a number.

Linear



r = 0.85 — and the correlation means what you think it means.

Curved



r ≈ 0 — yet the relationship is almost perfect. Correlation only measures **linear** association.

Outlier-driven



r = 0.8 — produced entirely by three points. Remove them and it vanishes.

In a consulting report, never present a correlation without the scatterplot that produced it. It is the difference between evidence and decoration.

Two More Tools: Lag Plots and Scatterplot Matrices

A lag plot is a scatterplot of a series against **its own past**. It is the visual bridge to autocorrelation, which is the whole of Session 3.

Lag plots • `gg_lag()`



Points hugging the diagonal at lag 12 means each month resembles the same month a year earlier — that is seasonality, seen as a relationship rather than a shape.

Scatterplot matrices

When several potential drivers exist, plot them all against each other at once. `GGally::ggpairs()` gives every pairwise scatterplot, every correlation and every distribution in one grid.

- Read the target row first: which candidate drivers actually move with your variable?
- Then read the rest: drivers strongly correlated with *each other* carry the same information twice.
- A curved pattern anywhere in the grid is a warning that a linear model will underperform.

```
sales |>
  as_tibble() |>
  select(units, temp, promo, holiday) |>
  GGally::ggpairs()
sales |> gg_lag(units, geom = "point")
```

EVERYTHING FROM TODAY, IN ONE PLACE

Session 2 R Cheat Sheet

```
# 0. build and check the tsibble
df |> as_tsibble(index = month,
                 key = store)
df |> has_gaps()
df |> fill_gaps()
# 1. time plot
df |> autoplot(units)
# 2. seasonal plot
df |> gg_season(units)
df |> gg_season(units, polar = TRUE)
```

```
# 3. seasonal subseries plot
df |> gg_subseries(units)
# 4. relationships
df |> gg_lag(units, geom = "point")
df |> as_tibble() |>
  select(units, temp) |>
  GGally::ggpairs()
# change the frequency
df |> index_by(week = ~ yearweek(.)) |>
  summarise(units = sum(units))
```

Five Ways These Plots Mislead

Each of these has cost a real forecasting project real money. Check for them before you show anything to a client.

Plotting sales as demand	When you stock out, sales are censored: the series shows what you sold, not what customers wanted. Forecast the censored series and you shrink demand every cycle.
Lines drawn across gaps	A missing month joined by a straight line invents data. Run `has_gaps()` before you plot, always.
Calendar effects mistaken for seasonality	February is short; some months have five weekends. Part of what looks seasonal is just an unequal number of trading days — adjust before concluding.
Aggregating away the pattern	Monthly totals hide a day-of-week pattern completely. Plot at the finest frequency you have before deciding the level to model.
Spurious correlation	Two trending series correlate almost by construction. Before believing a driver, ask whether the relationship survives once the shared trend is removed.

BLOCK 2 · 30 MINUTES

Business Case Lab

Take today’s four graphics to your own case. Working code is not the requirement — a decision you can defend is.

WHAT TO PRODUCE IN THE NEXT 30 MINUTES

Today's Lab Task

MINUTES 0–10

Get a series on screen

- Build or repair the tsibble: index, key, no gaps.
- Decide the frequency you will work at — and write down why.
- Produce one time plot with readable gridlines and labelled axes.

MINUTES 10–22

Interrogate the seasonality

- List every seasonal period you believe exists in your case.
- Run `gg_season()` and `gg_subseries()` for each one.
- Conclude: confirmed, rejected, or not enough data to tell.

MINUTES 22–30

Name one driver

- Pick the external variable you think matters most.
- Scatterplot it against your target — or state how you would obtain it.
- Answer the killer question: will you know its value at forecast time?

If your data does not exist yet, your output is a simulation plan: the patterns you will generate, the parameters, and the justification. That is a legitimate result for today.

STARTING POINT FOR EACH TEAM

Where to Point the Plots

CASE 1 · RETAIL

Happy Cow Ice Cream

Aggregate the POS transactions to daily cups and plot them. Then `gg_season(period = "week")` for the day-of-week pattern, and a scatterplot against Hong Kong daily temperature.

Watch for: seven months is under one annual cycle — you can see the weekly season, not the yearly one. Decide today how you will handle that.

CASE 2 · AIRLINES

Balearic Airlines

Plot twelve months of daily priority-class demand. Use `gg_season(period = "week")` for day-of-week effects and mark the holidays on the time plot.

Watch for: the National Day week. Does the holiday shift demand, or add to it? A subseries plot by weekday will show which weekdays carry the effect.

CASE 3 · MANUFACTURING

Clean Air Limited

Plot three years of monthly demand for the top items, then decide whether to plot each item, each supplier, or the total value. The choice of aggregation is your first real decision.

Watch for: the COVID collapse and the recovery surge. Mark them on the plot and decide now whether they belong in the estimation sample.

CASE 4 · E-COMMERCE

Wayfair

You are simulating: generate a small hierarchy of monthly series with the sparsity the article describes, then plot examples from each level.

Watch for: series with more zero months than non-zero ones. Your plots must make the sparsity visible — a line chart of mostly zeros is a bad graphic.

Before You Close the Laptop

Five lines in the decision log. This is what makes Session 3 start from somewhere instead of from scratch — and it is the raw material of your final report.

- **Frequency chosen** — and the reason. Daily, weekly, monthly?
- **Data problems found** — gaps, zeros, censoring, breaks, and what you did about each.
- **Seasonal periods** — which you tested, which the plots confirmed, which you rejected.
- **Transformation needed?** — does the variance grow with the level?
- **One driver** — the external variable you will pursue, and whether it is knowable in advance.

EXAMPLE ENTRY · SESSION 2

"Worked at daily frequency; weekly aggregation loses the weekend peak that drives staffing. Found 4 missing days in March (store closed, refurbishment) — kept as explicit gaps, not zeros. Weekly seasonality confirmed by `gg_season`; annual seasonality cannot be assessed with 7 months of data, so we will extend the series synthetically using temperature. Variance looks stable in absolute terms → no transformation for now. Driver: daily max temperature, available as a public forecast up to 7 days ahead, which matches our horizon."

BLOCK 3 · 20 MINUTES

Case Clinic

Two or three teams, five minutes each: what you tried, what you found, where you are stuck.

How the Clinic Runs

Five minutes is short on purpose. Show one plot, say what it told you, and name the thing that is blocking you. The room then has three minutes to challenge it.

The presenting team

- One plot on screen. Not three.
- One sentence on what it rules in or out for your model.
- One open question you genuinely do not know how to answer.

Everyone else

You have four different cases in the room, which means four different failure modes. Challenge with questions like:

- “Is that seasonality, or an unequal number of trading days?”
- “Would you know that driver at forecast time?”
- “How many complete cycles is that actually based on?”

Me

I close each one with feedback on the decision, not the code — whether the reasoning would survive a client asking “why?”

Every team presents at least three times across the term. Being stuck is a valid thing to present.

Getting a wrong idea killed in five minutes is the cheapest thing that will happen to your project all term.

3 SEPTEMBER → 23 SEPTEMBER

You Have Twenty Days Before Session 3

The longest gap in the course, and it is deliberate. Treat it as a consulting sprint: the team that arrives on 23 September with a real dataset and sharp questions will spend the rest of the term ahead of the one that arrives with the case still unread.

WEEK 1 · 4–10 SEP

Understand the business

Read the case properly, all of you. Agree the one-sentence brief. Find out how this sector forecasts today and what it costs them when they are wrong.

Output: the brief + the business context.

WEEK 2 · 11–17 SEP

Build the dataset

Assemble everything into one tsibble: case data, public sources, and whatever you must simulate. Document each assumption as you make it — you will not remember them in December.

Output: a tsibble that loads + the data inventory.

WEEK 3 · 18–22 SEP

Explore and challenge it

Produce today's four graphics on your own data and write what each rules in or out. Then list what you cannot explain — that list is what makes Session 3 worth attending.

Output: the plots + three concrete questions for me.

23 SEP · SESSION 3

Arrive ready

We add autocorrelation on top of what you have built and close the data-understanding phase. Session 3 is not a fresh start — it is the finish of this sprint.

Bring: laptop, tsibble loaded, questions written.

Three weeks of unsupervised team work is the most consulting-like thing in this course. Treat 23 September as a client meeting, because that is what it is.

SESSION 2 RECAP

What You Should Take Away

Plot before you model. Exploratory graphics are step 3 of the workflow and they close off most of the modelling space for free.

Four graphics, one line of code each. Time plot, seasonal plot, subseries plot, scatter/lag plot — they answer different questions.

Resolution is a design decision. Fit the axis to the data, choose gridlines a reader can count, and set an aspect ratio that shows the slope.

Correlation needs its scatterplot. Non-linearity, outliers and shared trends all produce numbers that lie.

Next session: we make this formal. Autocorrelation turns “I can see a pattern” into a number you can test — and white noise gives you the baseline for “there is nothing here”.

HOMEWORK

Before Session 3

Your case

- Finish the four plots you started in the lab and put them in one document.
- Write a one-paragraph “what the data looks like” summary — the first section of your final report.
- Log the five decisions from the checklist.

Method

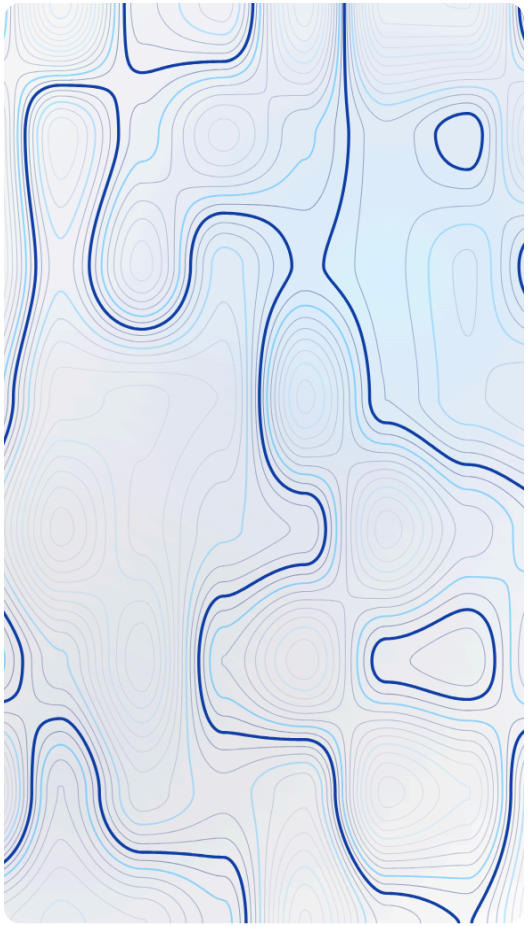
- Read **FPP3 chapter 2**, sections 2.2 to 2.7 — otexts.com/fpp3.
- Do exercises 2.1 to 2.5, using the built-in datasets.
- Skim 2.8 and 2.9 — lag plots and autocorrelation — as preparation for Session 3.

Come with

One plot from your case that you cannot explain. A pattern that should not be there, a break you cannot account for, or a seasonality that comes and goes.

We will use them as the worked examples in Session 3.

Session 3 · 23 September · Lags, autocorrelation and the ACF. Twenty days from now — use them.



END OF SESSION 02

TIME SERIES ANALYSIS

Time Series Graphics

NEXT · SESSION 3 · 23 SEPTEMBER

Lags, autocorrelation and the ACF

Twenty days from now. Use them: read the case, build the dataset, explore it, and bring three concrete questions.

No annex in this session. Everything you need is in the R cheat sheet on slide 24 and the sprint plan on slide 32.

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